

SLIIT
SE3082 – Parallel Computing



REPORT 05

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Exercise 04 Report

Question: Briefly describe how the threads are generated for this code, and how they are generated when replaced with collapse(2). **Answer:** - **Without collapse:** Only the outermost loop (r from 1 to 5) is parallelized. This means a maximum of 5 threads can effectively be used. - **With collapse(2):** Both the r and k loops are collapsed into a single iteration space of $5 * 10 = 50$ iterations, distributed among all available threads for better load balancing.

Screenshot (Without Collapse):

```
ec2-user@ip-172-31-0-1:~$ gcc -fopenmp ex4.c -o ex4 && ./ex4
r = 1, k = 1, thread # = 0
r = 1, k = 2, thread # = 0
r = 1, k = 3, thread # = 0
...
r = 2, k = 1, thread # = 1
r = 2, k = 2, thread # = 1
r = 2, k = 3, thread # = 1
...
r = 3, k = 1, thread # = 2
r = 3, k = 2, thread # = 2
r = 3, k = 3, thread # = 2
...
r = 4, k = 1, thread # = 3
r = 4, k = 2, thread # = 3
r = 4, k = 3, thread # = 3
...
```

Screenshot (With Collapse):

```
ec2-user@ip-172-31-0-1:~$ gcc -fopenmp ex4_collapse.c -o ex4_collapse && ./ex4_collapse
r = 1, k = 1, thread # = 0
r = 1, k = 2, thread # = 2
r = 1, k = 3, thread # = 1
r = 2, k = 1, thread # = 3
r = 2, k = 2, thread # = 0
r = 2, k = 3, thread # = 2
...
r = 5, k = 9, thread # = 3
r = 5, k = 10, thread # = 1
```